Spectral indices

Launcher: `Toolbox/spectral_indices.bat`

Purpose

The tool calculates a fixed catalogue of band arithmetic products. Vegetation and red-edge indices can enhance differences in crop vigour, chlorophyll, moisture response, or senescence that sometimes correspond to buried features. They also respond strongly to ordinary agronomic and environmental variation.

Inputs and parameters

Option	What to provide	Behaviour
Input image	An 8-band WV3 or LEGION GeoTIFF	The source bands are mapped by position using the selected sensor.
Results folder	Existing folder, or blank	Blank creates `SPECTRAL_INDICES` in the input image's grandparent directory. A manually selected folder must already exist.
Vector layer	Optional clip polygon	Crops every index output to the cutline.
Sensor	`WV3` or `LEGION`	Controls band mapping. Default: `WV3`. LEGION position 7 is RE2 and position 8 is N; WV3 position 7 is N and position 8 is N2.

Indices generated

The following definitions have all required bands for both sensors:

Group	Indices
NIR/red	BAI, DVI, IPVI, NDVI, NLI, OSAVI, RDVI, SAVI, SR, TDVI, TVI
NIR/green	GDVI, GNDVI, GRVI, NDSI
Multiband vegetation/chlorophyll	MTVI, RDRER, TCARI, TCARI/OSAVI, TCI, TriVI

LEGION can additionally generate **GM1** (`RE2/G`) and **TRRVI** because it has RE2. WV3 cannot generate those two. **NBLI** and **NDVIT** require a thermal band (`T1`), which neither mapping provides, so they are skipped.

The console prints each attempted index, its formula, and any missing band names. A skipped index is not necessarily a failed run.

Outputs

Each successful calculation produces:

- `<INDEX>_<input-name>.tif`

Products are Float32, DEFLATE-compressed GeoTIFFs and preserve the first source band's NoData value. When a cutline is supplied, the final product is a clipped Cloud Optimized GeoTIFF. Existing products are skipped; there is no overwrite control in the GUI.

Interpretation cautions

- Ratio indices can become extremely large or undefined where the denominator is zero or nearly zero. Inspect statistics and masks before display stretching.
- Soil background, crop species, growth stage, irrigation, disease, drainage, shadow, and recent cultivation can all create strong responses.
- Compare several complementary indices and, where possible, several acquisition dates. A persistent spatial pattern that also fits archaeological morphology is more persuasive than a single bright or dark index patch.
- Confirm the sensor choice and source band order. A plausible-looking result can still be scientifically wrong if bands were mapped incorrectly.
- Using images containing reflectance values in integer format, due to the application of a constant, can produce anomalous results for indices such as NBR.